

Halloween Math Crossword Solutions

R.I.P ALL Solutions Lie Here...

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ONE DOWN

This problem is a common example of a derangement – a permutation which leaves no element in its original place. In this case, we are looking for the probability that none of the jack-o-lanterns are being returned to their original owner. As the number of elements in set approaches infinity, the probability that a derangement happens approaches $1/e$. Thus, the answer is $100/e \approx 37$.

THREE ACROSS

NOTE: Cases excluded are not attainable while being the shortest distance

Case 1: Charlie hits houses A & C on the way to the party and B on the way back

$$(2C2) * (3C3) * (3C1) * (4C1) * (4C1 - 2) = 24$$

Case 2: Charlie hits houses A & B on the way to the party and C on the way back

$$(2C2) * (2C1) * (4C1) * (3C1) * (5C2 - 1) = 216$$

Case 3: Charlie hits house B on the way to the party and A & C on the way back

NOTE: This case is achievable by walking Case 1 in reverse. Thus, there are 24 routes.

Case 4: Charlie hits house C on the way to the party and A & B on the way back

NOTE: This case is achievable by walking Case 1 in reverse. Thus, there are 216 routes.

$$x - 404 = (24 + 216 + 24 + 216) - 404 = 76$$

ONE ACROSS

Let's assume that the number of candies that Sally originally has is x . The first statement tells us that $x \bmod 5 \equiv 4$. The second statement tells us that $x - 4 \bmod 4 \equiv 2$ which simplifies to $x \bmod 4 \equiv 2$. The third statement tells us that $x - 4 - 2 \bmod 3 \equiv 1$ which simplifies to $x \bmod 3 \equiv 1$.

OUR THREE CONDITIONS:

1. $x \bmod 5 \equiv 4$
2. $x \bmod 4 \equiv 2$
3. $x \bmod 3 \equiv 1$

Combining conditions 2 & 3, we deduct that $x \bmod 12 \equiv 10$. Listing integers, we see that 10, 22, **34** is our answer.

TWO DOWN

Legendre's formula gives that the exponent of p in the factorization of $n!$ is $\sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor$

While 28^2 , 27^2 , 26^2 , 25^2 , and 24^2 are the largest factors of $n!$, they all are comprised of smaller prime factors which are relatively easy to account for. However, 23^2 requires n to have 2 or more multiples of $23 \leq n$. The smallest n which satisfies this condition is **46**, making it our answer.

AUTHOR'S NOTE

We hope you guys enjoyed these questions! Please let us know if anyone discrepancies in problem solutions exist by messaging in the #math-help channel of our Discord:

<https://tinyurl.com/MEXplained>

1. 3	2. 4
3. 7	6

